CODECHECK certificate

Reviewed poster: De la Paz Ruiz, N., Augustijn, E.W., Farnaghi, M., Zurita-Milla, R. (2021): Population census estimation for Spatial Microsimulation.

To cite this report, use: Markus Konkol (2021): CODECHECK of *Population census estimation for Spatial Microsimulation*. DOI: 10.5281/zenodo.5543112

Summary: The reproduction was successful. The author followed good scientific practice and provided open code and open data via a GitHub repository. The two relevant figures from the poster could be reproduced with only minor design-related deviations from the figures reported in the poster, caused by a different aspect ratio. The author demonstrated concern for the reproducibility of the work by improving the code throughout the review process.

Date of check: 1st October, 2021

CODECHECK Repository: https://github.com/MarkusKonk/population.modelling.hue2010

CODECHECKER notes

I first forked the repository and opened the R Markdown file in RStudio. A first execution attempt resulted in an error because some of the libraries were not installed. All needed libraries were available in the package system. A second execution attempt resulted in errors because the data and the sourced R file could not be found. I changed the file paths from absolute to relative paths, which worked. ***Recommendation:*** *Use relative instead of absolute file paths*.

I also changed the file names, which included spaces. This can cause an issue when reading the file on a different system. ***Recommendation:*** *Use hyphens or underscores in file names. Spaces can cause issues on other systems.* After fixing these issues, I successfully created the two figures from the poster (see below). ***Recommendation:*** *Add the original poster or a separate output file that is not overwritten by executing the code. This allows others to compare the reproduced and original results.*

Overall, it was straightforward to reproduce the figures. The next level of reproducibility could be achieved by using the mybinder service (which would have the positive side effect of having a list of versioned libraries) and open data formats, e.g., *.csv* instead of *.xls*.

Reproduced Figures

Figure 3

The aspect ratio causes some design-related deviations. The difference in the correlation is likely because of a rounding error.

Original:

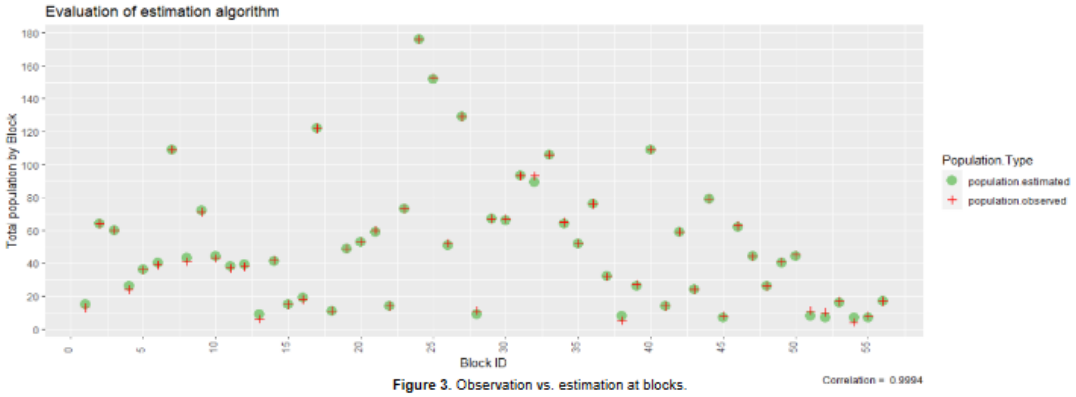


Figure 3. Observation vs. estimation at blocks.

Reproduced:

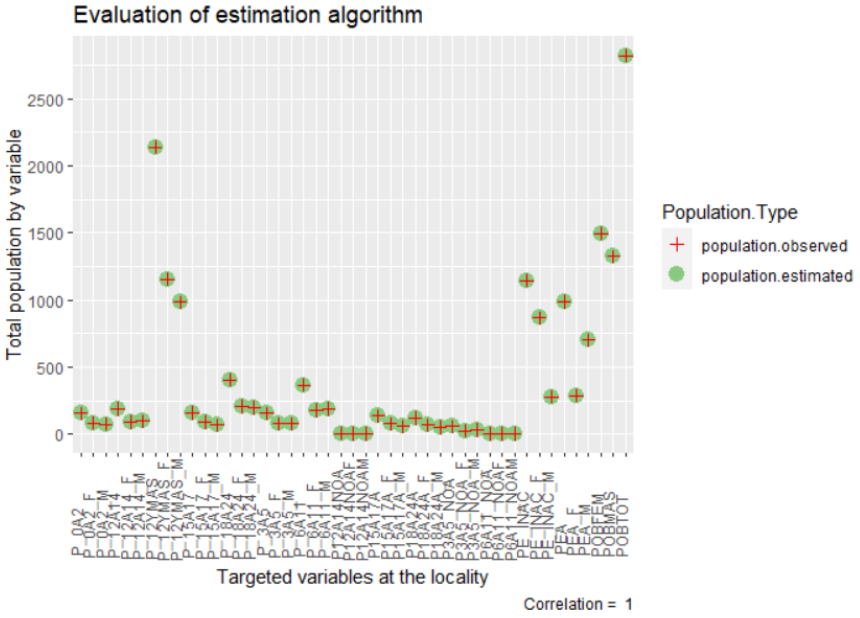


Figure 4

The aspect ratio causes some design-related deviations. The difference in the correlation is likely because of a rounding error.

Original:

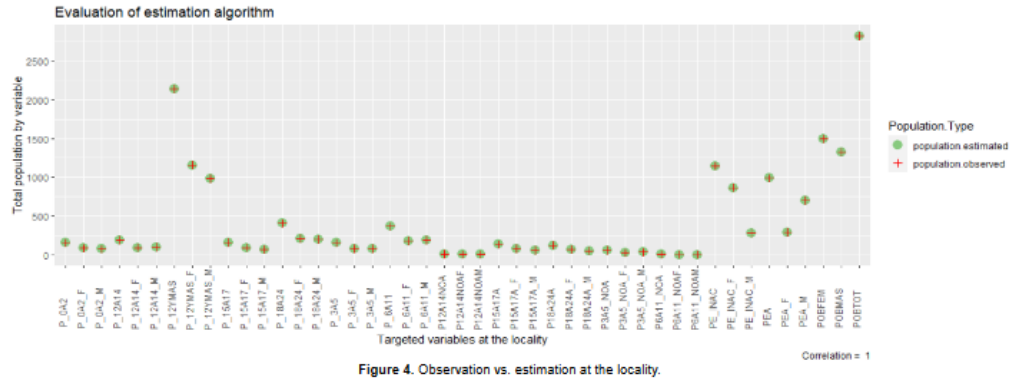
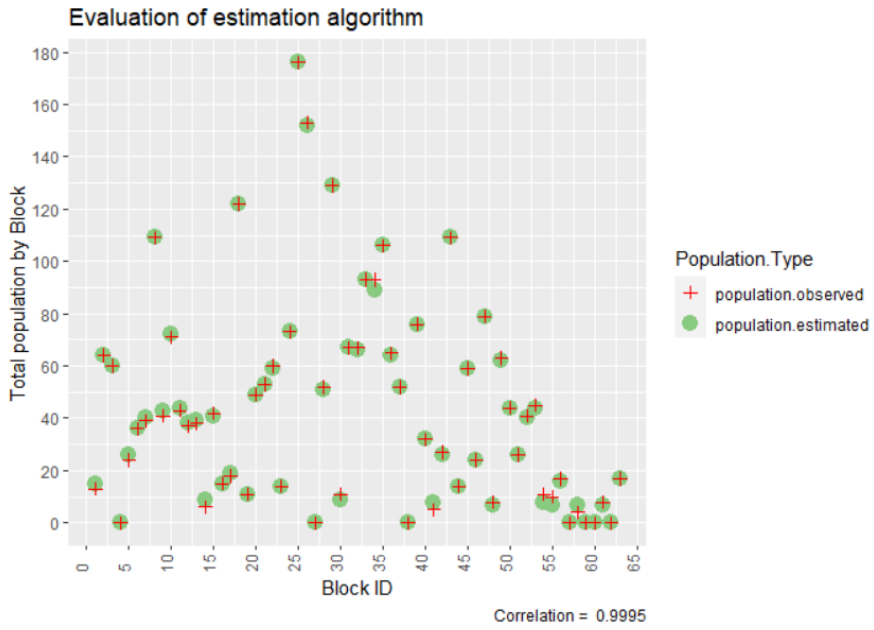


Figure 4. Observation vs. estimation at the locality.

Reproduced:



Used computational environment

Windows 10 Enterprise, Intel Core i7-8665U CPU 1.90GHz 2.11GHz, 32GB RAM, 64-bit Operating System. R version 4.0.3, RStudio Version 1.4.1103, rmarkdown_2.10, knitr_1.33